

MachEmbodied-U0: Unified Understanding and Generation Model for Embodied Intelligence

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Abstract

General-purpose robot control requires models to understand task intent, identify where to interact, capture how the scene evolves, and generate precise actions. Vision-language-action models provide strong semantic priors but typically do not explicitly model scene dynamics, while world-action models couple visual prediction with control without necessarily exposing the task-relevant semantic and spatial structure needed for fine-grained manipulation. We present **MachEmbodied-U0 (ME-U0)**, a unified embodied foundation model connecting understanding and generation experts through a Mixture-of-Transformers architecture. Subtask prediction and affordance grounding guide joint visual-dynamics and action generation via flow matching. Visual dynamics encompass future RGB, depth, surface normals, and optical flow, providing complementary supervision for appearance, geometry, and motion. Multi-rate Rotary Position Encoding (MRPE) aligns visual dynamics with fine-grained control. We pretrain ME-U0 on approximately 4,200 hours of curated demonstrations from robotic datasets and egocentric datasets. Using only the supervision natively available in each downstream benchmark, ME-U0 achieves an average score of 17.66 on the RoboDojo simulation benchmark and average success rates of 99.0% and 82.5% on LIBERO and LIBERO-Plus, respectively. We additionally validate ME-U0 on real-world robotic manipulation tasks, demonstrating its effectiveness beyond simulation. Without corresponding downstream supervision, ME-U0 further demonstrates zero-shot subtask prediction, affordance grounding, and visual dynamics on simulated and real-world observations. Overall, ME-U0 combines competitive downstream control performance with transferable task-grounding and visual-dynamics capabilities across simulation and the real world.

Project Page: <https://machembodied.com/ME-U/ME-U0.html>

Github: <https://github.com/MachEmbodied/ME-U0>

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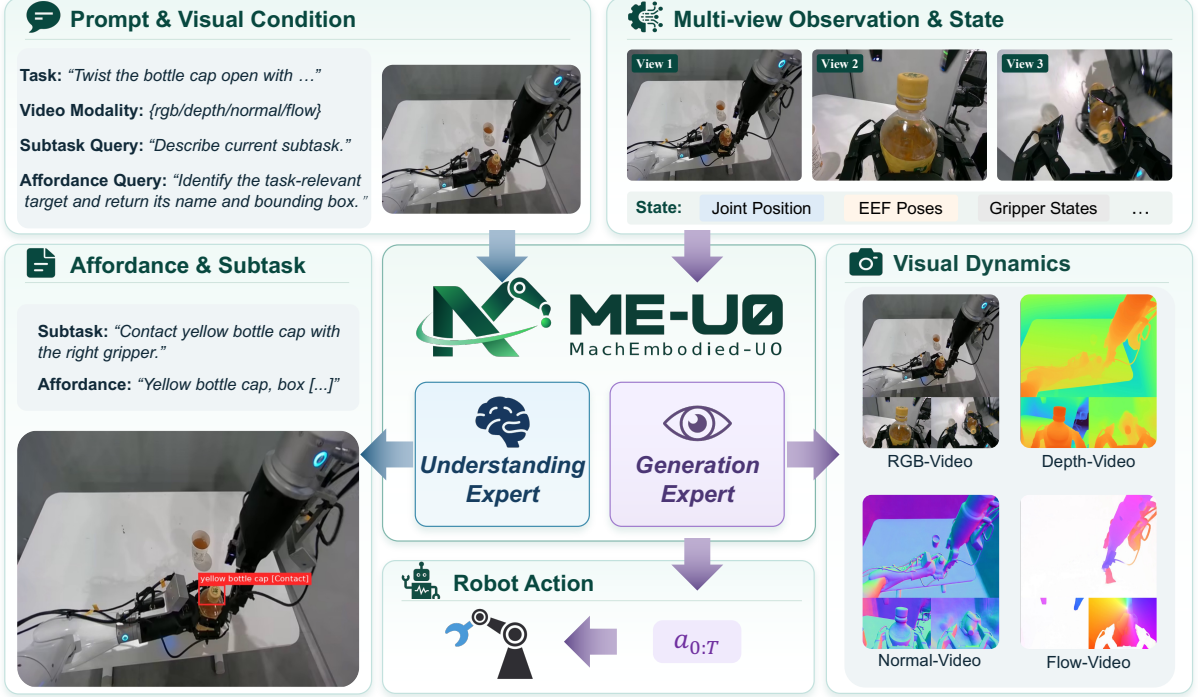


Figure 1. MachEmbodyed-U0 (ME-U0) integrates task-grounded understanding with joint visual and action generation through cooperating understanding and generation experts.

1 Introduction

Physical intelligence requires more than recognizing objects or following instructions: a robot must determine what to do, where to interact, and how its actions will change the world. These requirements connect semantic understanding, prediction, and control. A general embodied foundation model should therefore ground task intent in the observed scene while relating executable actions to their anticipated physical consequences.

Vision-language-action (VLA) models leverage the semantic priors of pretrained vision-language models for language-conditioned robot control [1–9]. However, their action-centric objectives typically do not explicitly capture how the scene evolves during interaction. World-action models (WAMs) address this limitation by jointly modeling future observations and robot actions, providing dense supervision for spatial structure and motion [10–14]. Yet visual dynamics alone does not ensure task-relevant grounding, interaction localization, or long-horizon task decomposition. These complementary strengths motivate a model that connects semantic understanding and visual dynamics with continuous control.

Unified multimodal architectures offer a natural basis for this integration [15–17]. Recent embodied extensions explore several forms of unification: UAM [18] separates semantic and visuomotor pathways to preserve multimodal competence during action learning; Motubrain [19] jointly models language-conditioned video and actions across policy, world-modeling, and inverse-dynamics modes; and Cosmos 3 [20] scales unified reasoning and generation across language, vision, audio, and actions.

Together, these advances show that semantic reasoning, visual dynamics, and control can coexist within a unified architecture. They also expose a remaining design question: interleaving outputs, separating expert pathways, or supporting flexible modality configurations does not by itself determine how task-level intent should be translated into spatially grounded and dynamically informed control. For manipulation, the central challenge is therefore not merely to generate text, visual observations, and actions, but to organize

their interaction around four coupled questions: what operation should be performed next, where the interaction should occur, what visual, geometric, and motion changes should unfold during execution, and how the operation should be realized through continuous control. This motivates a manipulation-centered form of unification that connects task decomposition and affordance grounding with geometry- and motion-aware visual dynamics and action generation.

We introduce **MachEmbodied-U0 (ME-U0)**, a unified embodied foundation model capable of subtask decomposition, scene understanding, visual dynamics generation, and continuous action generation. As illustrated in Fig. 1, the model couples an understanding expert with a generation expert in a Mixture-of-Transformers (MoT) architecture. Conditioned on task instructions, visual observations, and robot states, the two experts connect scene-grounded task interpretation with visual dynamics generation and continuous control. Specialized parameters accommodate the distinct demands of semantic understanding and continuous generation, while shared multimodal attention enables coordination between them. ME-U0 combines three complementary designs: explicit subtask reasoning and affordance grounding, joint multimodal visual-dynamics and action generation, and action-substep temporal alignment. Together, these designs establish a direct pathway from high-level intent and local interaction geometry to anticipated scene evolution and executable control.

The understanding expert predicts subtasks and affordances to specify *what to do next* and *where to act*. Subtask prediction connects the overall instruction to the current manipulation stage, while affordance prediction identifies and localizes the relevant interaction target in the observed scene. Together, these outputs translate high-level intent into actionable semantic context. Through shared multimodal attention, the generation expert can use this context to condition future visual and action predictions, integrating task understanding into the control process.

Conditioned on this semantic context, the generation expert jointly predicts visual dynamics and continuous actions through Flow Matching [21], coupling control with anticipated scene changes in a shared backbone. We extend RGB generation to depth, surface normals, and optical flow [22], adding explicit geometry and motion supervision to support precise manipulation. MRPE aligns action steps with temporally compressed visual transitions.

To scale pretraining across heterogeneous robot embodiments, we map embodiment-specific states and actions into a canonical interface while preserving their physical control semantics. Dataset-specific transformations align joint, gripper, base, torso, and head variables, while validity masks indicate the dimensions and time steps available for each sample. This interface enables a single model to learn from approximately 4,200 hours of curated robotic and egocentric demonstrations, with the robotic corpus spanning four datasets and six embodiments. Through this pretraining, ME-U0 jointly acquires task-grounded understanding, geometry- and motion-aware visual dynamics, and continuous control.

For downstream evaluation, we use only the supervision natively available in each benchmark, without introducing additional subtask, affordance, geometry, or motion annotations during post-training. ME-U0 achieves a score of 17.66 on RoboDojo and average success rates of 99.0% and 82.5% on LIBERO and LIBERO-Plus, respectively. These results demonstrate that the representations acquired during large-scale pretraining transfer effectively to downstream manipulation, even under reduced-supervision post-training.

Our contributions are fourfold:

- **A unified embodied foundation model.** We present ME-U0, which integrates task-grounded understanding, geometry- and motion-aware visual dynamics, and continuous robot control within a shared Mixture-of-Transformers architecture.

- **Grounded visual–action dynamics modeling.** Subtask prediction and affordance grounding provide structured semantic and spatial context for generation. Future RGB, depth, surface normals, optical flow, and continuous actions are jointly modeled through flow matching, while MRPE aligns fine-grained control with temporally compressed visual dynamics.
- **A scalable embodied pretraining system.** We construct a heterogeneous pretraining mixture with unified robot state–action semantics, multimodal annotations, task-aware sampling, and efficient data infrastructure, enabling training across four datasets and six robot embodiments.
- **Broad empirical validation.** ME-U0 achieves strong downstream performance on RoboDojo, LIBERO, and LIBERO-Plus, and further demonstrates effective transfer to real-world robotic manipulation.

2 Related Work

2.1 Vision-Language-Action Models

Vision-language-action (VLA) models ground visual observations and language instructions in robot actions. RT-1 established the benefits of large-scale, multi-task robot learning, while RT-2 incorporated pretrained vision-language knowledge into control through action-token prediction [4, 23]. Open generalist policies further improve accessibility and adaptation: Octo accommodates new observation and action spaces, and OpenVLA supports efficient fine-tuning of a pretrained vision-language backbone for manipulation [5, 6].

Another line of work focuses on action representation and generalization. FAST compresses action sequences using frequency-domain tokenization, whereas π_0 uses flow matching to generate continuous actions, offering different approaches to modeling dexterous control [7, 9]. Building on π_0 , $\pi_{0.5}$ combines data from multiple robots and the web with high-level semantic prediction, enabling long-horizon manipulation in previously unseen homes [8]. Together, these methods highlight the complementary roles of transferable semantic knowledge, diverse training data, and expressive action representations in generalist robot policies.

2.2 World Action Models

World action models (WAMs) connect visual dynamics modeling with action learning, using future prediction to inform control or shape policy representations. Early approaches include UniPi, which extracts actions from language-conditioned video plans, and GR-1, which jointly predicts future images and robot actions after video generative pre-training [10, 11]. DreamZero scales joint video–action modeling with a pretrained video diffusion backbone, supporting zero-shot policy generalization and transfer from video-only demonstrations across embodiments [12].

Recent methods examine how much future generation is necessary for effective control. LaWAM predicts compact latent visual subgoals to condition action generation, reducing the cost of reconstructing future video. Fast-WAM retains video co-training but omits future prediction at inference, showing that predictive supervision can benefit policies without requiring explicit test-time imagination [13, 14]. These approaches expose an important design choice: whether to use predicted futures as inference-time context or primarily as supervision for learning representations of scene dynamics.

2.3 Unified Models

Unified multimodal models combine understanding and generation through shared multimodal context. Show-o integrates autoregressive language modeling with discrete diffusion for visual generation, while BAGEL scales unified pre-training on interleaved multimodal data [15, 16]. Lance further explores shared context with separate expert pathways for image and video understanding, generation, and editing [17]. These models provide foundations for coupling semantic reasoning with visual prediction.

For embodied control, BagelVLA interleaves language planning, visual forecasting, and action generation to support long-horizon manipulation, while Cosmos 3 extends a unified mixture-of-transformers architecture to language, image, video, audio, and action sequences [20, 24]. UAM addresses multimodal forgetting during action training by introducing a parallel Dorsal Expert initialized from a generative model and supervised to predict visual dynamics [18]. Its separation of semantic and control-related processing complements joint generation approaches, emphasizing that integrating perception, prediction, and action also requires preserving pretrained understanding capabilities.

3 Training Data

We draw on a raw pool of approximately 5,700 hours of robotic demonstrations and 3,920 hours of egocentric demonstrations. Egocentric data complements robotic data by covering manipulation contexts and behaviors underrepresented in robot demonstrations. After careful curation, we retain approximately 4,200 hours of demonstrations for pretraining. We further annotate a subset of the data with subtask, affordance, geometry, and motion labels to support joint understanding and generation.

3.1 Dataset Composition

3.1.1 Robotic Datasets

Our robotic data comes from four open-source datasets: AgiBot World Beta [25], RoboMIND 2.0 [26], RoboCOIN [27], and Galaxea Open-World [28]. After validity checks, the resulting corpus spans six robot embodiments and a broad range of manipulation tasks. Fig. 2(a) summarizes its dataset and embodiment composition.

Beyond data sources and embodiments, we construct a *scene–action–object* tag scheme. The three dimensions describe the scene context of a task, the behaviors it involves, and the objects it mentions. For visualization clarity, we aggregate related tags into six scene groups, six action groups, and eight object groups. Fig. 2(b) summarizes scene–action coverage. Common behaviors recur across different task contexts, while each context encompasses multiple forms of manipulation. This overlap places shared skills in varied settings, extending the mixture beyond a narrow set of context-specific behaviors. Fig. 2(c) complements this view with action–object coverage, showing how shared behaviors are associated with objects of different physical properties and functional roles.

The data mixture connects a broad range of task meanings with concrete robot interactions across diverse scenes and objects, providing rich supervision for grounding semantic understanding in physical behavior and supporting transfer across tasks.

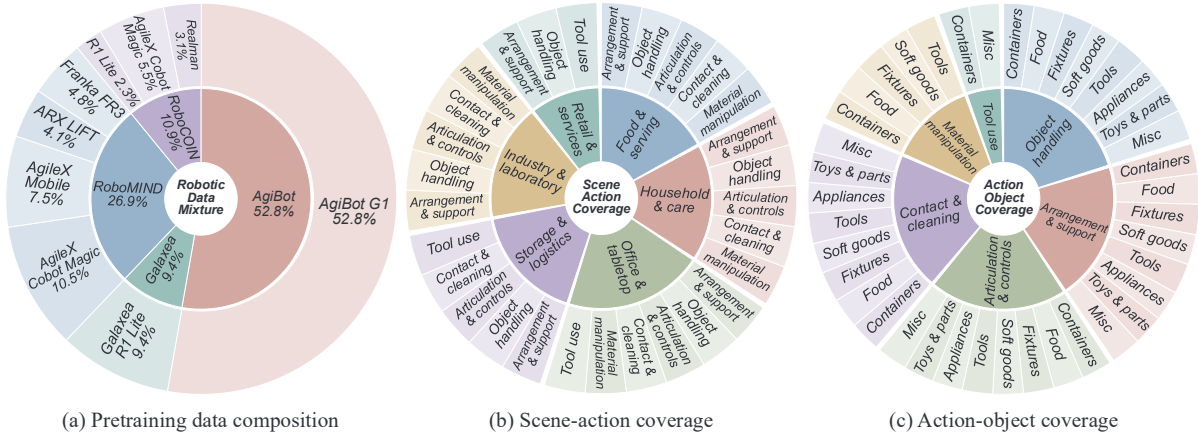


Figure 2. Robotic data composition and scene-action-object diversity. (a) Dataset and robot-embodiment composition. (b) Task descriptions and the actions in each task. (c) Actions and the objects in task descriptions.

3.1.2 Egocentric Datasets

To improve coverage of task tags underrepresented in the robotic datasets, we select egocentric demonstrations with matching tags and incorporate them into the overall pretraining mixture. For action-supervised pretraining, we primarily use EgoDex [29], EgoLive [30, 31], EgoVerse [32], HOI4D [33], HOT3D [34], and the hand-annotated portion of Ego-Exo-4D [35]. We select demonstrations with temporally aligned first-person observations and low-level human motion annotations, including fingertip locations, wrist poses, and arm or shoulder states. These annotations support conversion into the unified hand-centric action representation described in Section 3.3.

3.2 Robotic Data Processing

3.2.1 Data Curation

Fig. 3 illustrates our three-stage cleaning pipeline and representative signal anomalies. Following recent robot-data curation pipelines [36, 37], we adopt dataset-specific filtering policies to account for differences in state definitions, units, sampling rates, and action conventions. Before filtering, source-specific readers establish the physical meaning, units, ordering, and temporal alignment of the recorded channels. Each source is evaluated using checks supported by its available signals.

We apply three checks in sequence. First, we detect sudden changes in continuous state and action channels after scaling each dimension by its $q_{99} - q_{01}$ range. A median filter followed by smoothing provides a reference trajectory; large residuals, accelerations, or jerks flag an episode. Second, we assess state-action consistency by estimating the command-response lag from cross-correlations of smoothed first differences, then measuring directional agreement at lag-aligned timesteps where actions vary. Low agreement across the configured number of dimensions, or a state freeze during action variation, flags the episode. Third, we compare raw values with manually reviewed, per-dimension physical bounds informed by full-dataset

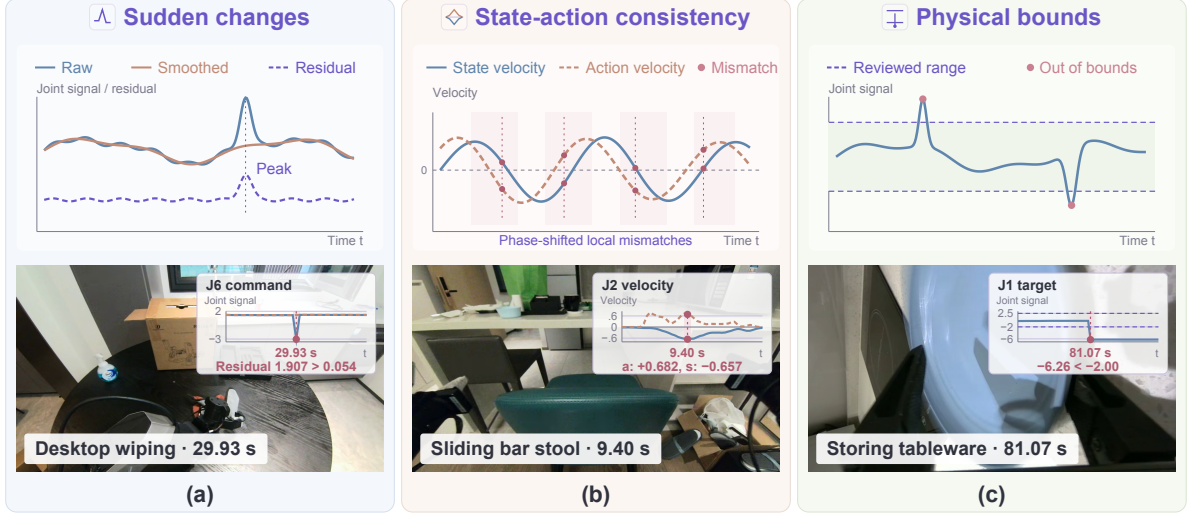


Figure 3. Schematic of real-robot data curation. Each stage removes a subset of the episodes passed to it.

statistics. Gripper channels are excluded from these continuous-signal checks because their open/close values do not represent smooth motion. An episode is excluded if any enabled check is triggered or a source-specific timestamp-gap rule is violated. Filtering operates on complete episodes and leaves the original recordings unchanged, preserving the temporal continuity of retained demonstrations.

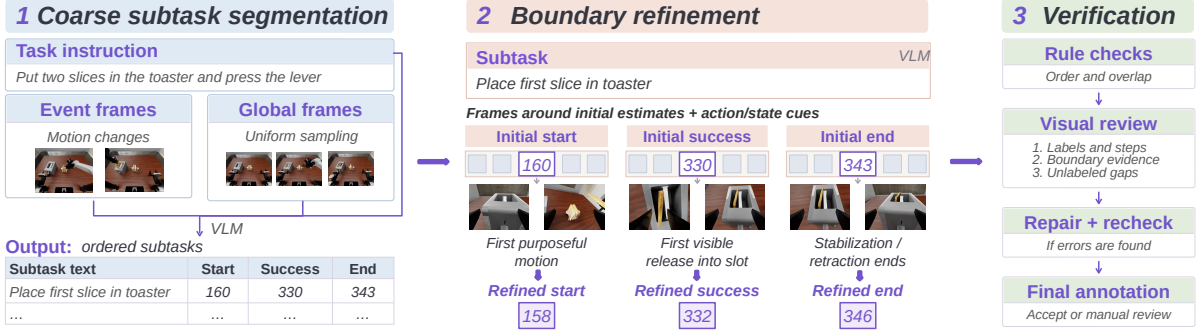
3.2.2 Data Annotation

We augment selected portions of the data with task-grounded semantic, spatial, geometric, and motion annotations.

Subtask Labeling. We develop a three-stage pipeline for labeling robot demonstrations with temporally grounded subtasks shown in Fig. 4. Our annotation pipeline comprises three stages. *Coarse subtask segmentation* combines event timestamps identified from action and state changes with uniformly sampled timestamps, capturing both salient transitions and the overall task progression. Using synchronized multi-view observations at these timestamps and the task instruction, the pipeline identifies an ordered sequence of subtasks and their approximate temporal intervals. *Boundary refinement* then examines visual observations near the proposed timestamps, together with temporally aligned action and state changes, to refine each subtask’s start and end boundaries and identify the first frame where its goal is visibly achieved. *Verification* checks the resulting sequence for semantic coherence, temporal ordering, and boundary consistency. Annotations that fail verification receive one automatic revision, with unresolved cases referred for manual review. We use GPT-5.5 [38] for all stages.

Affordance Labeling. Affordance annotations identify the target object and interaction location at each manipulation stage, thereby bridging high-level task semantics and low-level robot actions. As illustrated in Fig. 4, the annotation pipeline comprises three stages. *Candidate frame proposal* uses a rule-based procedure adapted from AffordanceVLA [39] to extract candidate frames from each demonstration. *Keyframe selection* jointly examines the head-camera video, task instruction, subtask timeline, and candidate contact sheet. It selects frames depicting meaningful interactions and assigns each a target category and an affordance instruction. The *Spatial grounding* stage localizes the target object and its interaction point in each selected frame. Frames are upsampled to twice their original resolution for finer localization; predicted coordinates are then mapped back to the original image space and checked for spatial consistency. The model we use is Qwen3.6-35B-A3B [40].

Subtask annotation



Affordance annotation

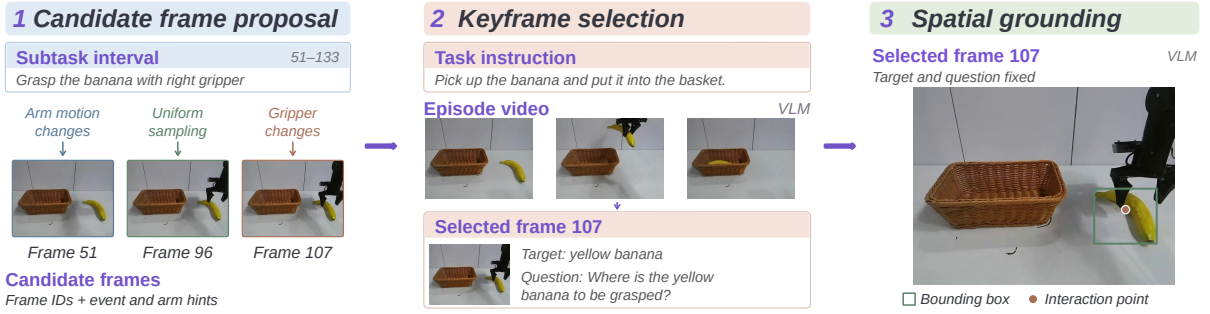


Figure 4. Overview of our data annotation pipeline. We annotate temporal subtasks through segmentation, boundary refinement, and verification, and obtain spatial affordances through keyframe selection and grounding.

Visual Dynamics Labeling. We derive depth, surface normals, and optical flow labels from the original RGB data to provide complementary geometric and motion supervision for our unified model. These auxiliary annotations are intended to strengthen the model’s spatial and temporal understanding, supporting its ability to reason about scene structure and motion in robotic manipulation.

We use MoGe-2 [41] to estimate metric depth and surface normals from individual RGB frames, and normalize the depth labels [22]. To capture motion across frames, we use WAFT [42] to estimate optical flow between sampled frame pairs. We adjust the temporal stride for each dataset according to its sampling frequency, since the same frame offset can correspond to different time intervals across datasets. Together, these annotations provide supervision for both scene geometry within individual frames and visual motion across time.

3.3 Egocentric Data Processing

Annotation Standardization. Action-annotated Egocentric demonstrations are standardized into a unified human-centric action format following LeRobot v3 [43], with invalid frames and discontinuous action chunks filtered before task-level normalization. Each action-supervised sample is centered on an anchor timestep and contains first-person observations, future action chunks, available language supervision, and hand states represented in end-effector space. Hand states are represented by 3D position and axis-angle orientation, providing a consistent state-action space across heterogeneous human trajectories.

Ego2Robot Synthesis. As illustrated in Fig. 5, we synthesize robot aligned demonstrations through task understanding, scene reconstruction, trajectory optimization, and rendering to reduce embodiment mismatch and depth uncertainty while improving contact accuracy.

We first use Qwen3.6 [40] to parse each egocentric video and its instruction, identifying relevant objects,

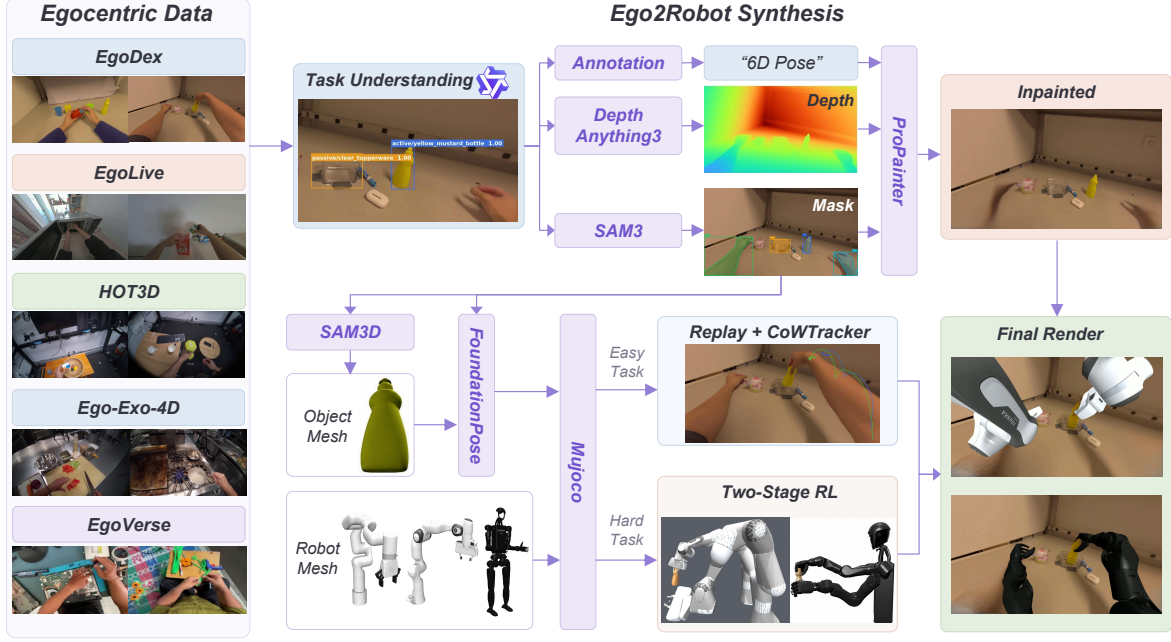


Figure 5. Overview of egocentric data processing. Ego-to-Robot synthesis converts human demonstrations into robot-aligned observations and trajectory annotations.

interaction roles, and physical properties. Based on these attributes, each episode is categorized as either an easy task suited to trajectory replay or a hard task requiring contact refinement. For both task categories, the scene is reconstructed using metric depth from Depth Anything 3 [44] and masks from SAM3 [45]. Following Inpaint Anything [46], we then apply ProPainter [47] to the arm and hand masks to remove human regions and recover the occluded background for all episodes.

Trajectory processing differs between the two task categories. For easy tasks, we replay the annotated trajectories and use CoWTracker [48] to ensure their accuracy in pixel space, thereby reducing the 2D-3D mismatch. For hard tasks involving contact sensitive interactions, we extend prior work [49–52] to recover object geometry and motion. Manipulated rigid objects are reconstructed as 3D physical models by using SAM3D [53], while masks, depth, and 6D pose estimates [54, 55] are fused to recover temporally consistent object trajectories. The corresponding wrist and fingertip trajectories are then refined through a two-stage Proximal Policy Optimization (PPO) [56] curriculum to improve contact accuracy and trajectory smoothness. The optimal trajectories from both branches are subsequently retargeted to the embodiment equipped with grippers through robot-specific inverse kinematics, solved with Mink [57] in MuJoCo [58]. Finally, the robot mesh is rendered into the inpainted egocentric video, with occlusions resolved according to object depth. The rendered images and optimized trajectory annotations constitute the final robot-aligned dataset.

4 Model Design

4.1 Main Architecture

MachEmbodied-U0 integrates an understanding expert and a generation expert within a unified multimodal framework, as illustrated in Fig. 6. The understanding expert processes language instructions and visual content to predict subtasks and affordances, while the generation expert predicts future visual observations

ME-U0 Architecture

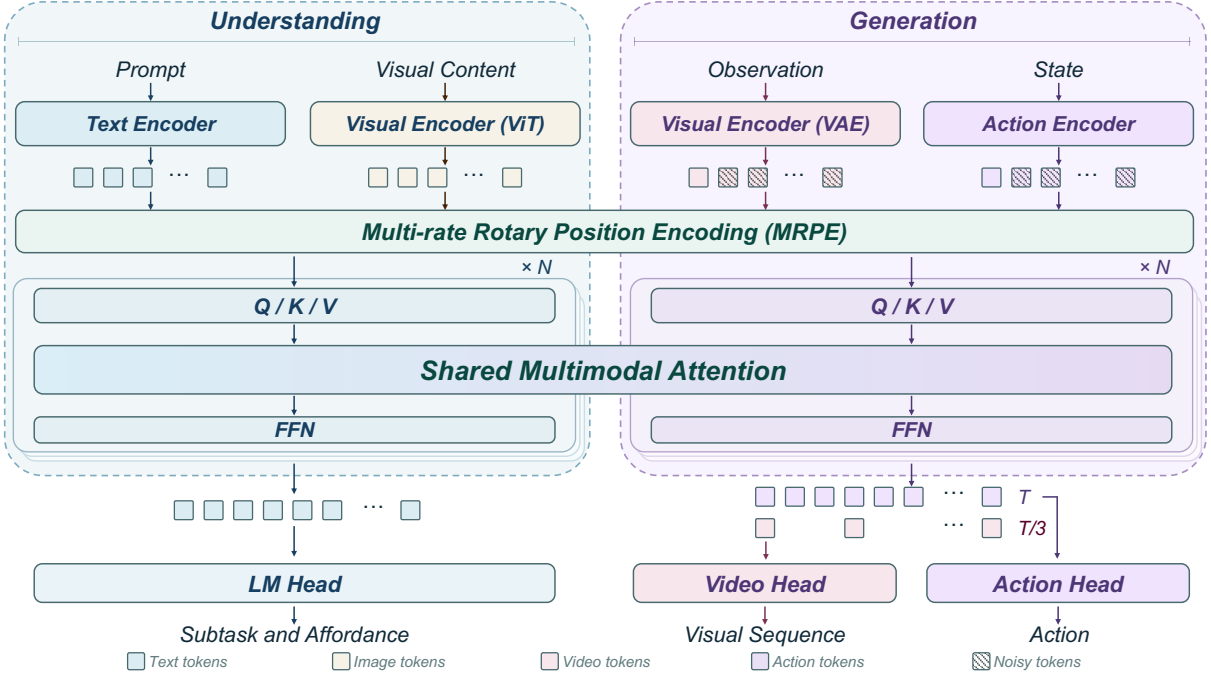


Figure 6. Overview of our framework. The understanding expert predicts subtasks and affordances, while the generation expert jointly predicts future visual observations and robot actions through shared multimodal attention.

and robot actions. The two branches are coupled through a standard Mixture-of-Transformers (MoT) architecture. We initialize MachEmbodied-U0 from Lance [17] and further train it to acquire robot-specific capabilities.

4.1.1 Understanding Expert

We formulate subtask and affordance prediction as autoregressive text generation and train the understanding expert with a next-token cross-entropy objective. The resulting semantic visual tokens are combined with the task instruction in the shared multimodal sequence.

Subtask Prediction. An overall task instruction can remain unchanged across several manipulation stages, while the immediate goal changes as the scene evolves. Following the use of semantic subtask supervision in $\pi_{0.5}$ [8], we ask the understanding expert to describe the local manipulation goal given the instruction and current observation. This encourages the expert to interpret the observed scene in the context of the overall task and distinguish the behaviors required at different stages.

Affordance Prediction. For keyframes with affordance annotations, we model actionable spatial grounding using a similar question-answering formulation. The **Affordance Query** asks the model to identify a task-relevant target in the current image and predict its bounding box, followed by the intended interaction and the corresponding interaction point. When both annotations are available, the affordance follows the subtask in the same assistant response; otherwise, it is predicted alone. Samples without affordance annotations retain their original training objective.

4.1.2 Generation Expert

As shown in Fig. 6, the generation expert jointly generates future visual dynamics and robot actions. In the shared sequence, text tokens follow the current observation and robot-state tokens and precede the future

visual and action tokens. Causal attention prevents text tokens from attending to future targets. During denoising, the generation expert uses shared multimodal attention to access representations of the input images and text from the understanding expert. It generates RGB, depth, surface-normal, or optical-flow sequences together with continuous robot actions.

Visual Dynamics Generation. The generation expert produces visual sequences in the modality specified by the input prompt: RGB, depth, surface normals, or optical flow. We represent geometry and motion as three-channel visual targets [22], complementing RGB appearance with scene structure, surface orientation, and inter-frame motion. All modalities share a generation backbone and latent output projection operating in the latent space of a common video VAE, whose decoder reconstructs the predicted latents into the requested visual modality. This formulation unifies RGB, geometry, and motion generation without modality-specific prediction heads.

Action Generation. For action generation, we represent each timestep in the predicted action sequence as one token. At each denoising step, the generation expert processes the noisy action tokens together with the visual tokens through bidirectional self-attention. An action-specific output head maps the resulting action representations to velocity predictions, which are used to update the noisy action sequence using explicit Euler steps. This process proceeds in synchrony with visual generation to produce the final action sequence.

4.2 Multi-rate Rotary Position Encoding

We adopt the multimodal rotary position embedding (mRoPE) scheme [3], which encodes text positions and visual spatiotemporal coordinates through rotations of query and key representations. Building on this scheme, we introduce Multi-rate Rotary Position Encoding (MRPE) to accommodate temporally subsampled video while retaining actions at their original sampling rate. Each video latent frame corresponds to a chunk of consecutive actions, whose internal order would remain unspecified if they shared identical position indices. We therefore retain the shared temporal position of the corresponding video latent frame, but assign each action a distinct height-axis index according to its order within the chunk, while keeping the width-axis index fixed. This preserves temporal alignment between actions and video while explicitly encoding the position of each action within its chunk.

4.3 Unified Multimodal Prompting

Inspired by the use of contextual information in the prompts of $\pi_{0.7}$ [59], we design a unified multimodal prompt that combines the current visual observation and task instruction with control metadata and explicit prediction requirements. Control-mode and action-semantics fields specify the robot’s control conventions, including relative or absolute commands and gripper encoding. Subtask and affordance queries define the requested understanding outputs, while a video-modality field specifies whether to generate RGB, depth, surface normals, or optical flow. Together, these fields provide the task and control context for joint understanding and generation. An example is shown below, with the shared system prompt omitted and visual embeddings represented by an observation placeholder:

Table 1. Unified state and action representation. Dimensions are specified per arm or gripper.

Component	Dim.	Shared representation
Arm: joint control	6 / 7	Joint configuration in radians
Arm: EEf control	9	Cartesian position in metres and 6D orientation
Gripper	1	Normalized closing coordinate: 0 = open, 1 = closed
Chassis	3	Body-frame planar velocity (v_x, v_y, ω_z), in m/s and rad/s
Torso: AgiBot	2	Pitch and lift position, in radians and metres
Torso: Galaxea	3	Planar Cartesian velocity (v_x, v_z, ω_y), in m/s and rad/s
Head	2	Yaw and pitch joint positions in radians

```

<Current RGB observation> Task:pick up the cup. Video Modality:rgb. Control
Mode:joint. Action Semantics:joint delta (chunk-start); gripper absolute (0=open,
1=closed; pre-norm). Subtask Query:Describe the current subtask based on the task
and the current image. Affordance Query:Identify a target relevant to carrying
out the task in the current image. Give its name and bounding box.
Output:<subtask><affordance>

```

4.4 Unified Action Representation

Embodiment diversity provides rich manipulation experience, but inconsistent control conventions can obscure the common structure across demonstrations. Similar numerical values may describe different physical quantities or even opposite control directions. We therefore adopt a unified action space for robot states and actions across robot embodiments, shown in Tab. 1.

We implement dataset-specific adapters to align the physical meaning of robot states and actions before statistical normalization. These adapters reconcile units, coordinate conventions, and control directions while preserving meaningful embodiment-specific differences. For example, gripper states and actions are expressed in a shared closing coordinate, where 0 denotes fully open and 1 fully closed. This reverses and rescales Galaxea’s native $[0, 100]$ commands while preserving AgiBot’s action convention. AgiBot’s millimetre-valued gripper states are calibrated separately to the same closing coordinate.

Some sources additionally require kinematic conversion to make states and actions physically consistent. Galaxea, for instance, records mobile-base states as wheel-steering angles and wheel velocities, but specifies actions as Cartesian velocities. We convert these wheel-level states into body-frame planar velocities, giving states and actions a common physical interpretation. Controls with distinct physical meanings, such as torso position and velocity, remain explicitly distinguished.

5 Pretraining

5.1 Training Data Sampling

Broad coverage of tasks and scene contexts provides a foundation for learning manipulation skills that generalizes beyond the pretraining data [8, 60]. However, diversity in the collected corpus does not necessarily translate into diversity in training exposure. Under a fixed training budget, sampling in proportion to recorded timesteps can concentrate supervision on abundant tasks and long demonstrations, limiting exposure to less common behaviors. An effective sampling strategy should therefore preserve the breadth of the task repertoire while providing meaningful supervision across tasks.

Table 2. Source-specific video–action temporal alignment. H_a denotes the number of action steps, and H_v the number of sampled video frames.

Dataset source	Action (Hz)	Video (Hz)	H_a / H_v	Horizon (s)
AgiBot World Beta	30	10	36 / 13	1.20
Galaxea Open World	15	5	24 / 9	1.60
RoboMIND 2.0: AgileX / Mobile / ARK	30	10	36 / 13	1.20
RoboMIND 2.0: Franka	15	5	24 / 9	1.60
RoboCOIN: 30 Hz recordings	30	10	36 / 13	1.20
RoboCOIN: 50 Hz recordings	25	8.33	36 / 13	1.44

We adopt global task-aware sampling to achieve this, allocating the training budget across source-specific task buckets rather than prescribing dataset-level proportions. Each eligible task receives an initial coverage allocation drawn from multiple episodes, ensuring that less abundant tasks also contribute to pretraining. With this coverage established, the remaining budget is then distributed through temperature-based sampling with weights

$$w_k = \max(R_k, 1)^\alpha,$$

where R_k denotes the task’s remaining capacity in training blocks and α controls the degree of distribution smoothing. We set $\alpha = 0.5$ to balance the contribution of less abundant tasks against the greater demonstration diversity available in larger task buckets. This sublinear weighting reduces the dominance of large task buckets while retaining a preference for tasks with more available data. Task allocations remain bounded by the available data capacity, avoiding aggressive oversampling of tasks with few demonstrations. For tasks with insufficient capacity, we draw additional samples from egocentric datasets to supplement their coverage. This combines broad task coverage with more balanced supervision without enforcing equal sample counts across tasks. Dataset proportions emerge naturally from the resulting task-level allocation. Using this strategy, we draw 120 million training samples from the eligible tasks in the curated data mixture.

5.2 Video–Action Temporal Alignment

Heterogeneous recording rates introduce a temporal mismatch when combining robotic datasets: sequences with the same number of frames may capture substantially different amounts of motion and span different physical durations. We address this by jointly selecting source-specific sampling strides and sequence lengths, keeping prediction windows within approximately 1–2 seconds while sampling actions three times as densely as video. This provides finer-grained control supervision without requiring equally dense visual sequences.

Specifically, each interval between consecutive sampled video frames corresponds to three action steps. For H_a action steps and H_v video frames, including the anchor, we have

$$H_a = 3(H_v - 1).$$

Combined with the video VAE temporal compression, this establishes a consistent correspondence of 12 action steps per predicted video-latent step. The resulting representation preserves a common cross-modal temporal structure across datasets while accommodating their different recording rates. Tab. 2 summarizes the source-specific sampling settings.

5.3 Understanding Expert Training

We introduce two complementary language-supervision tasks to enhance task-grounded understanding. Subtask prediction connects the overall goal to the ongoing manipulation step, while affordance prediction identifies and localizes a task-relevant object or region in the current image. Together, they encourage the expert to associate semantic intent with concrete visual referents, providing richer supervision than continuous action targets alone. Both responses are trained with autoregressive cross-entropy

$$\mathcal{L}_{\text{und}} = -\frac{1}{T} \sum_{t=1}^T \log p_{\theta}(y_t \mid y_{<t}, \mathbf{x}), \quad (1)$$

where $\mathbf{x}$ comprises the current image, task instruction, and contextual metadata, and $\mathbf{y} = (y_1, \dots, y_T)$ is the target response containing the available subtask and affordance labels. The loss is evaluated only on supervised response tokens, jointly learning task interpretation and visual grounding within a shared language-generation framework.

5.4 Generation Expert Training

The generation expert, initialized from the pretrained Lance [17] model, jointly denoises future visual latents and robot actions using conditional Flow Matching [21]. Given the conditioning context $\mathbf{c}$, we corrupt the clean visual target $\mathbf{z}$ and action trajectory $\mathbf{a}$ with independent standard Gaussian noise at a shared flow time τ . We sample $\tau = \text{sigmoid}(u + \log 4)$, where $u \sim \mathcal{N}(0, 1)$, yielding a shifted logit-normal distribution that emphasizes higher noise levels:

$$\mathbf{z}_{\tau} = (1 - \tau)\mathbf{z} + \tau\epsilon_z, \quad \mathbf{a}_{\tau} = (1 - \tau)\mathbf{a} + \tau\epsilon_a. \quad (2)$$

The noisy visual and action tokens are processed together by the generation expert, allowing information exchange between modalities. Separate output projections predict their respective velocity fields:

$$(\hat{\mathbf{u}}_z, \hat{\mathbf{u}}_a) = v_{\theta}(\mathbf{z}_{\tau}, \mathbf{a}_{\tau}, \tau; \mathbf{c}). \quad (3)$$

The joint flow-matching objective is

$$\mathcal{L}_{\text{gen}} = \mathbb{E} \left[\lambda_z \|\hat{\mathbf{u}}_z - (\epsilon_z - \mathbf{z})\|_{\mathbf{M}_z}^2 + \lambda_a \|\hat{\mathbf{u}}_a - (\epsilon_a - \mathbf{a})\|_{\mathbf{M}_a}^2 \right], \quad (4)$$

where $\|\cdot\|_{\mathbf{M}}^2$ denotes the mean squared error over valid elements selected by mask $\mathbf{M}$.

We adopt joint denoising to preserve the unified understanding and generation architecture of our model. Language and visual understanding provide the conditioning context, while the same generation expert jointly processes noisy action tokens and future visual latents. Introducing a separate action denoiser would decouple action generation from this shared generative process and weaken the architectural unification of understanding, visual prediction, and robot control. Instead, joint denoising incorporates actions directly into the multimodal generation process, allowing action and visual predictions to interact throughout denoising. Modality-specific output projections accommodate their different representations while preserving a shared generation backbone.

For visual dynamics generation, selected future RGB targets are replaced with depth, surface normals, or optical flow, with the target modality specified by the language instruction. All modalities share the same video VAE and generation expert, while the current RGB observation and action targets remain

unchanged. This allows geometric prediction and action generation to be jointly optimized under the same flow-matching objective.

Beyond joint video and action generation, the architecture supports training on forward and inverse dynamics tasks [20] by varying the conditioning inputs and prediction targets. Forward dynamics predicts future visual latents conditioned on the current observation and an action trajectory, whereas inverse dynamics predicts actions conditioned on the current and future visual latents:

$$\text{Forward dynamics: } (\mathbf{c}, \mathbf{a}) \longrightarrow \mathbf{z}, \quad (5)$$

$$\text{Inverse dynamics: } (\mathbf{c}, \mathbf{z}) \longrightarrow \mathbf{a}, \quad (6)$$

$$\text{Joint policy prediction: } \mathbf{c} \longrightarrow (\mathbf{z}, \mathbf{a}). \quad (7)$$

The overall training objective combines the generation expert’s joint flow-matching loss with the understanding expert’s autoregressive cross-entropy loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{gen}} + \lambda_{\text{und}} \mathcal{L}_{\text{und}}, \quad (8)$$

where λ_{und} controls the relative contribution of understanding supervision, while λ_z and λ_a in $\mathcal{L}_{\text{gen}}$ balance visual and action generation. This objective jointly optimizes task-grounded understanding, future visual prediction, and robot action generation.

6 Training Infrastructure

To train efficiently on our large-scale pretraining data mixture, we design our training infrastructure to address two costly recurring operations: dataset initialization and video data loading. The data-initialization optimizations reduce the initialization time of the full pretraining mixture from approximately one hour to less than ten minutes, and the video data loading strategy reduces overall training time by more than 30% in our training platform.

6.1 Efficient Dataset Initialization

Precomputed episode manifests. Initializing a large dataset mixture can be slow because it requires traversing source directories and inspecting episode files, particularly when metadata must be retrieved over the network. We move this discovery process to the data-preparation stage and record episode identifiers, lengths, task associations, and source-specific filtering information in persistent manifests. At training time, the episode catalog is reconstructed directly from these manifests, eliminating repeated file-system inspection.

Reusable Subtask Index. We precompute a reusable metadata index containing each subtask’s episode identifier, temporal boundaries, and language label. Training runs load this index directly, avoiding repeated parsing and alignment of the original annotations during initialization. Rather than storing experiment-specific training windows, the index preserves the complete annotated intervals, from which eligible anchors are derived according to the configured video and action offsets. It can therefore be reused across experiments with different prediction horizons.

Lightweight Data Readers. Constructing a full Dataset object for every constituent dataset introduces substantial initialization overhead, particularly for large sources distributed across hundreds of LeRobot

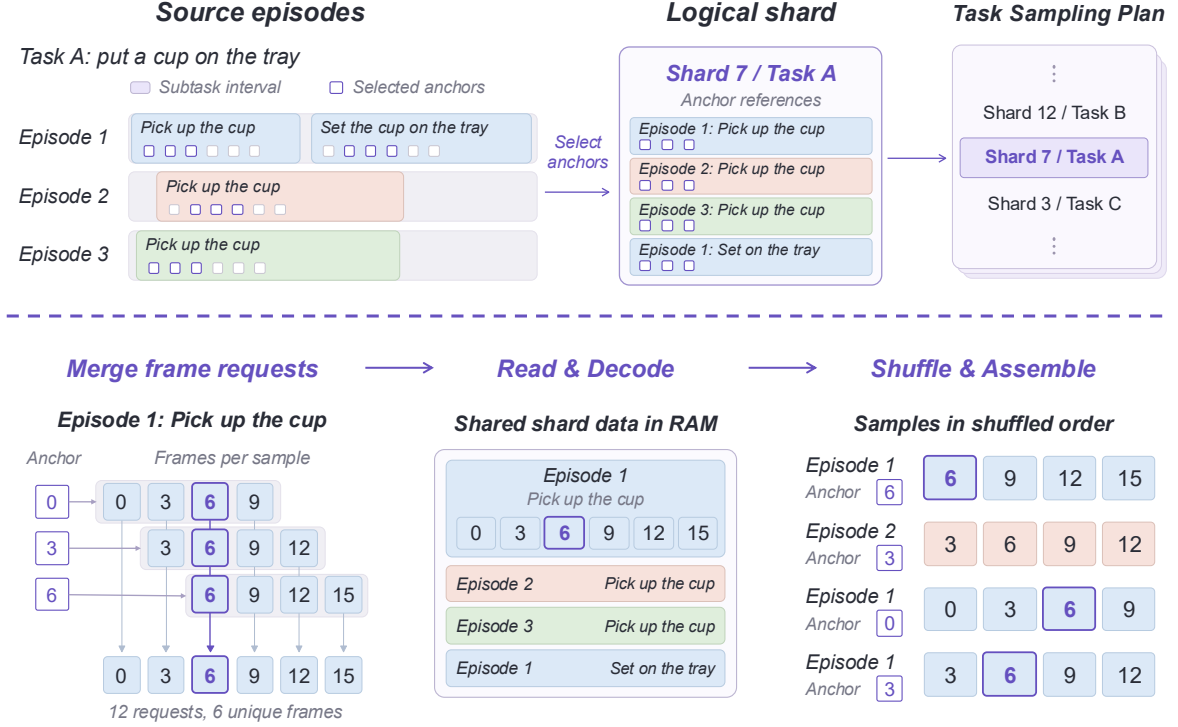


Figure 7. Shard-based sampling and loading. Top: selected anchors from subtask intervals across episodes of the same task are referenced by a logical shard from the global task sampling plan. Bottom: overlapping frame requests are merged, decoded into shared memory buffers, and assembled into samples in shuffled anchor order.

directories. To reduce this overhead, we implement lightweight LeRobot readers that use precomputed metadata to access the underlying files without repeating the full initialization procedure. We additionally support direct access to native data formats through source-specific readers, avoiding the cost of converting large datasets into a common storage format.

6.2 Logical Shards

Video-based model training incurs substantial I/O and video decoding overhead for the data pipeline. We organize training samples into logical shards, which serve as the basic units of allocation for the global task sampler and of grouped loading for data workers. This preserves temporal locality for efficient video decoding while allowing the sampler to control coverage across tasks and episodes.

The upper part of Fig. 7 shows how demonstrations are organized into logical shards. Each training sample starts at an *anchor*, whose complete video/action window must remain within the same subtask interval and episode. To construct a shard, we cycle through episodes of one task, taking nearby anchors from each subtask interval until the shard contains the configured number of anchors. These shards form the units of the global task sampling plan. Following the task-level allocation described above, the plan selects shards without replacement and shuffles them into a reproducible schedule for distributed workers.

The lower part of Fig. 7 shows how a worker loads a scheduled shard. Rather than loading each sample independently, a worker merges overlapping frame requests from nearby anchors within the same subtask interval. In the example, twelve requests reduce to six unique frames, which are decoded and kept in memory alongside the state/action data. The worker then shuffles the anchors and assembles their samples from the shared data. Training samples are thus drawn from different episode segments in shuffled order, while overlapping samples continue to share the same decoded frames.

Table 3. Results on RoboDojo-Sim. SR and Score are reported on a 0–100 scale, with higher values better. Bold indicates the best result within each model group.

Model	Gen-Std		Gen-Rand		Precision		Long-Horizon		Memory		Open		Overall	
	SR	Score	SR	Score	SR	Score	SR	Score	SR	Score	SR	Score	SR	Score
VLA														
StarVLA [61]	4.67	7.54	0.00	0.33	4.33	9.90	6.50	14.15	2.44	3.34	0.58	0.67	3.24	6.40
X-VLA [62]	12.22	17.90	1.33	3.04	12.00	18.32	9.75	16.53	3.56	4.76	0.50	0.55	6.52	10.13
$\pi_{0.5}$ [8]	14.89	20.93	1.44	5.82	5.50	12.40	14.67	23.54	4.67	5.89	1.67	1.98	6.93	11.44
Spatial Forcing [63]	14.89	21.25	3.78	6.98	10.58	17.32	14.58	23.26	4.11	5.43	1.58	1.78	8.04	12.38
Hy-Embodied-0.5-VLA [64]	16.56	21.98	0.22	1.57	8.00	13.81	14.92	25.74	12.11	13.37	0.58	0.65	8.80	13.07
Xiaomi-Robotics-1 [65]	28.00	35.65	6.00	11.44	18.83	26.69	23.67	38.39	6.56	7.81	3.58	3.94	13.93	20.07
GalaxeaVLA (G0.5) [66]	21.44	27.72	4.22	9.20	20.42	28.25	32.25	44.12	7.33	8.61	1.58	1.73	14.88	20.23
DM0.5 [67]	17.89	23.49	4.00	8.06	16.75	24.82	19.50	33.70	47.44	47.74	2.08	2.43	19.34	24.90
WAM														
Fast-WAM [14]	2.11	4.33	0.11	0.34	0.00	1.96	5.17	9.14	3.44	3.55	0.42	0.42	2.03	3.48
AHA-WAM [68]	6.22	10.32	0.33	1.26	2.42	5.86	2.67	8.61	2.78	2.97	0.83	0.88	2.39	4.82
GigaWorld-Policy-0 [69]	5.78	10.28	0.00	0.41	1.83	6.15	8.92	15.51	2.22	3.46	0.50	0.54	3.27	6.20
X-WAM [70]	5.33	11.24	1.33	3.54	1.83	6.72	9.08	17.47	4.67	6.32	0.25	0.57	3.83	7.69
OpenWAM- α [71]	25.56	33.16	4.11	8.26	9.25	18.45	25.33	34.93	9.11	10.41	1.08	1.41	11.92	17.18
ME-U0 (ours)	19.11	28.10	1.33	6.96	15.42	23.95	22.33	36.98	7.00	8.42	0.92	1.41	11.18	17.66

6.3 Video Decoding and CPU Offloading

We cap the number of nearby anchors processed per decoding job, restricting its video span to those anchors and their required future frames rather than the entire episode. This balances frame reuse against per-job memory demand. A separate concurrency limit controls aggregate CPU and memory usage, while asynchronous preparation of the next shard overlaps decoding with training.

To accommodate decoding workloads that exceed the CPU capacity of GPU nodes, we also support offloading data preparation to separate CPU jobs. These jobs read and decode data ahead of training and place the prepared payloads in a shared cache. Training workers retrieve cached payloads without performing local video decoding on cache hits. The CPU jobs follow the training sample schedule and maintain a bounded lead over consumption, preventing excessive accumulation of prepared data. This separation allows decoding capacity to scale independently of GPU resources and reduces training time significantly.

7 Experimental Results

7.1 Post-Training Recipe

We separately post-train ME-U0 on RoboDojo [72] and LIBERO [73], initializing both runs from the same pretrained checkpoint. Both runs use 368 PPU810E accelerators with a per-device batch size of 2. We use an action horizon of 24 with end-effector pose control for LIBERO, and an action horizon of 48 with delta joint-position control for RoboDojo.

It is worth noting that neither RoboDojo nor LIBERO provides subtask or affordance annotations, or auxiliary supervision for video geometry and motion. For fair comparisons, we do not augment either dataset with these additional supervision signals during post-training. Despite the potential degradation of pretrained representations under this reduced-supervision post-training setting, ME-U0 achieves strong

Table 4. Post-training results on standard LIBERO.

Model	Spatial	Object	Goal	Long	Avg.
VLA					
$\pi_{0.5}$ [8]	98.8	98.2	98.0	92.4	96.9
OpenVLA-OFT [74]	97.6	98.4	97.9	94.5	97.1
StarVLA [61]	97.8	98.6	96.2	93.8	96.6
ABot-M0 [75]	98.8	99.8	99.0	96.6	98.6
X-VLA [62]	98.2	98.6	97.8	97.6	98.1
WAM					
Fast-WAM [14]	98.2	100.0	97.0	95.2	97.6
Motus [76]	96.8	99.8	96.6	97.6	97.7
ImageWAM [77]	97.2	99.2	98.8	98.4	98.4
LingBot-VA [78]	98.5	99.6	97.2	98.5	98.5
OpenWAM- α [71]	99.6	99.6	99.8	98.2	99.3
ME-U0 (ours)	98.4	99.8	99.0	98.6	99.0

Table 5. Robustness on LIBERO-Plus. Success rates (%) are reported across seven perturbation categories [79].

Model	Camera	Robot	Language	Light	Background	Noise	Layout	Avg.
VLA								
StarVLA [61]	52.5	49.8	88.5	95.7	95.7	73.0	76.9	74.1
ABot-M0 [75]	60.4	67.9	86.4	96.2	91.6	86.4	82.6	80.5
$\pi_{0.5}$ [8]	78.4	73.6	80.8	96.2	94.1	89.0	84.5	84.4
Qwen-RobotManip [37]	87.2	75.5	85.6	96.6	97.7	97.7	87.3	89.0
WAM								
Fast-WAM [14]	16.4	44.5	68.9	78.2	53.7	37.7	60.7	51.5
Being-H0.7 [80]	82.0	59.0	82.8	97.8	90.0	93.5	88.5	82.1
Cosmos-Policy [81]	75.8	63.3	81.7	96.5	88.9	92.7	82.2	82.2
ImageWAM [77]	80.8	50.3	91.4	98.1	85.5	93.8	80.5	83.1
OpenWAM- α [71]	33.8	76.1	88.0	97.0	87.1	39.8	77.5	69.2
ME-U0 (ours)	70.2	75.7	90.4	97.6	81.9	81.3	84.5	82.5

performance on both benchmarks, highlighting the robustness of its pretrained representations and their effectiveness for downstream adaptation.

7.2 Simulation Benchmark Evaluation

7.2.1 RoboDojo

We evaluate ME-U0 on the 42 tasks of RoboDojo [72] across generalization, precision, long-horizon execution, memory, and open-vocabulary instruction following. For each of three evaluation seeds, we run 25 standard and 25 randomized episodes for each of the 12 generalization tasks, and 50 episodes for each of the remaining 30 tasks. We report success rate (SR) and the benchmark’s process score, which gives partial credit for task progress. Overall results average the five capability dimensions equally, with Gen-Std and Gen-Rand first combined into the generalization dimension.

Tab. 3 compares ME-U0 with representative VLA and WAM baselines. ME-U0 achieves the strongest overall performance among the evaluated WAMs, reaching an overall process score of 17.66 and an SR of 11.18%. The advantage is particularly evident in precision and long-horizon execution, where ME-U0

obtains process scores of 23.95 and 36.98, respectively. These results suggest that grounding joint visual-action dynamics with subtask and affordance context supports both fine-grained interaction and extended task execution. ME-U0 also remains competitive with established VLA baselines despite being pretrained on only approximately 4,200 hours of curated robotic and egocentric demonstrations, demonstrating competitive downstream transfer from a comparatively moderate pretraining corpus. Memory-dependent tasks remain comparatively challenging, with a process score of 8.42 and an SR of 7.00%, likely because the current model does not explicitly retain observation history. Incorporating temporal context and memory mechanisms could further improve performance when task-relevant information is no longer visible in the current observation.

7.2.2 Libero/Libero-Plus

We evaluate a single LIBERO-adapted ME-U0 checkpoint on the four standard LIBERO suites and LIBERO-Plus, which extends the same task families along seven controlled perturbation dimensions [79]. Standard LIBERO evaluates 50 initial states for each of its 40 tasks, resulting in 2,000 episodes, whereas LIBERO-Plus evaluates one episode for each of 10,030 perturbed task instances. We use the same checkpoint for both benchmarks without LIBERO-Plus-specific adaptation and aggregate success over episodes. The overall LIBERO-Plus result is therefore weighted by the number of task instances in each perturbation category. Published baseline results are transcribed from the cited reports for reference and were not rerun in our evaluation environment.

As Tab. 4 shows, ME-U0 achieves an average success rate of 99.0% on LIBERO, including 98.4% on Spatial, 99.8% on Object, 99.0% on Goal, and 98.6% on Long, demonstrating consistently near-ceiling performance across all four suites. Without additional adaptation, the same checkpoint achieves an overall success rate of 82.5% on LIBERO-Plus, as reported in Tab. 5. Performance remains particularly strong under lighting and language perturbations, with success rates of 97.6% and 90.4%, respectively. Camera-viewpoint changes and variations in robot initial states are the most challenging categories, yielding 70.2% and 75.7%. Success rates under background textures, sensor noise, and object layouts are 81.9%, 81.3%, and 84.5%, respectively. These results show that LIBERO-Plus provides a more discriminative assessment than the nearly saturated standard benchmark and identify robustness to viewpoint changes and robot initial states as areas for further improvement.

7.3 Real-World Deployment

We evaluate ME-U0 through direct deployment on real-world manipulation tasks across multiple robot platforms, with model inference performed on a remote server. The pretrained checkpoint is used without additional post-training or deployment-specific adaptation. Since these platforms are represented in the pretraining corpus, this evaluation examines deployment on familiar embodiments rather than generalization to unseen robot platforms. Fig. 8 presents three representative instruction-guided tasks: picking up a marker, placing a tissue into a box, and positioning the number 20 to complete the equation $3 + 17 = 20$. Spanning grasping, pick-and-place manipulation, and semantically grounded object placement, these demonstrations provide qualitative evidence of ME-U0’s ability to translate task instructions into executable actions in real-world settings.

7.4 Zero-Shot Analysis

To assess the generalization capabilities of ME-U0, we conduct three zero-shot experiments using its pretrained weights directly, without any additional post-training: visual dynamics generation, affordance

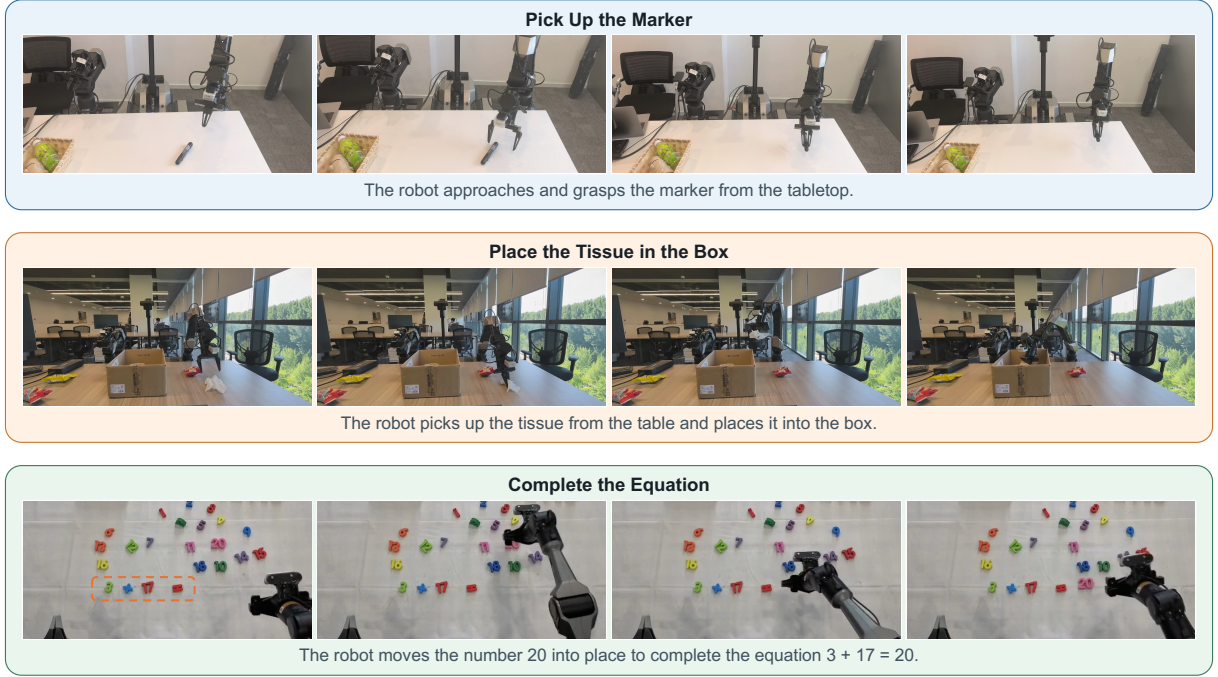


Figure 8. Real-world deployment demonstration of ME-U0.

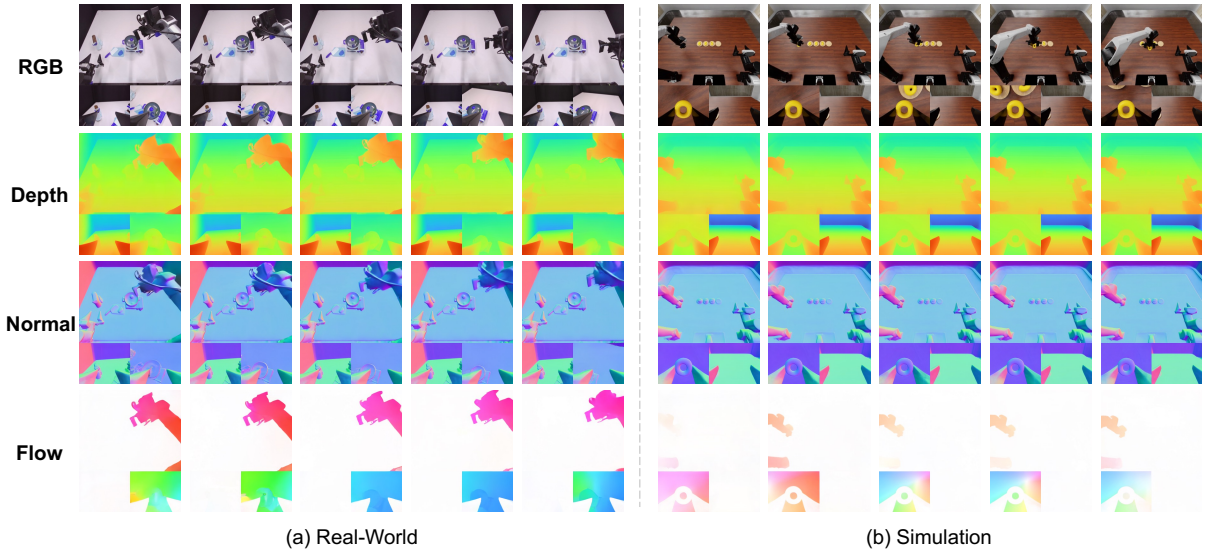


Figure 9. Zero-shot visual dynamics generation on RoboDojo (a) Real-World and (b) Simulation data. Each panel shows RGB observations alongside generated depth, surface normals, and optical flow at five time steps.

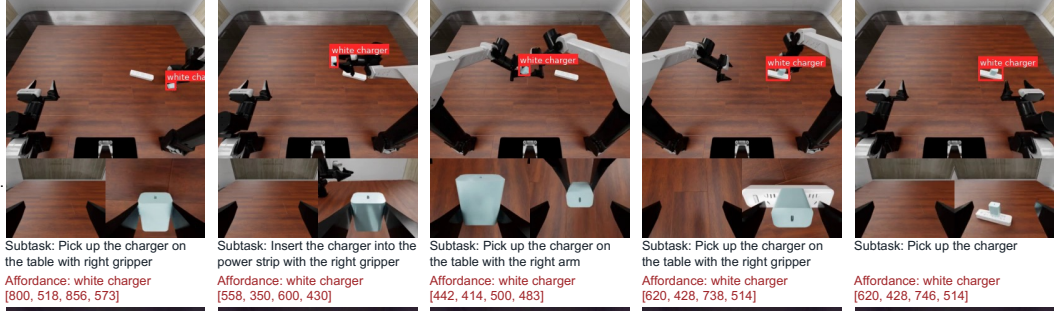
prediction, and subtask prediction. All evaluation samples are drawn from data excluded from our pretraining corpus and are used solely for evaluation.

7.4.1 Visual Dynamics

We evaluate visual dynamics generation on samples drawn from the simulation and real-world training splits of RoboDojo [72]. Neither split overlaps with our pretraining corpus, and the samples are used exclusively for evaluation without benchmark-specific fine-tuning. Fig. 9 shows qualitative results for both domains. ME-U0 generates depth maps, surface normals, and optical flow that capture meaningful scene structure and motion across real-world and simulated observations, demonstrating transfer to previously unseen data. Qualitatively, predictions on real-world samples are slightly more consistent than those on

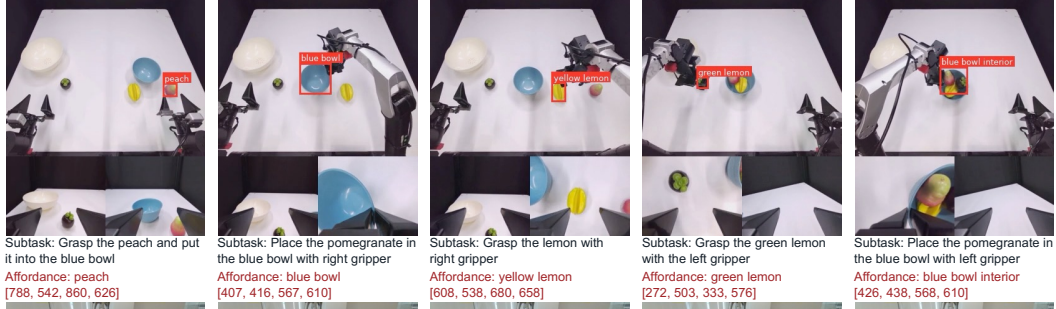
RoboDojo Simulation

Task:
Plug the charger into the power strip.



RoboDojo Real-World

Task:
Place the fruits into the blue bowl.



Self Collected Real-World

Task:
Pick up pen with left hand.



Public Datasets Real-World

Task:
Pick up the straw and insert it into the medium coffee cup.

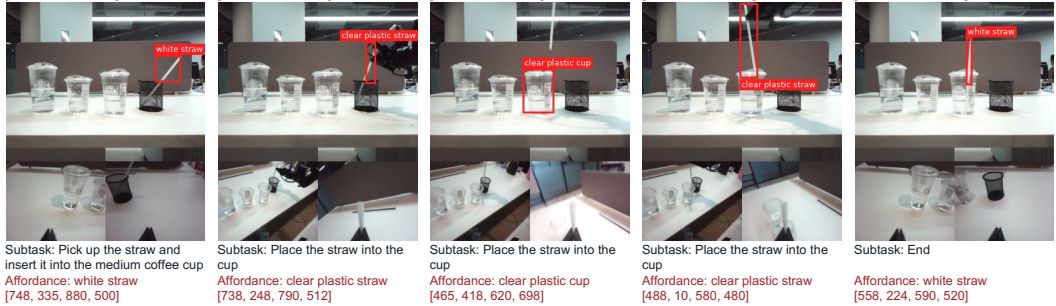


Figure 10. Zero-shot affordance and subtask prediction. Rows show RoboDojo simulation, RoboDojo real-world, self-collected, and RDT-1B [82] data. Task instructions appear on the left, with predicted subtasks and affordances below each image. Red boxes mark predicted task-relevant targets.

simulated samples. We hypothesize that this difference arises from the domain gap between simulated imagery and the real-world observations encountered during pretraining.

7.4.2 Affordance

We evaluate affordance prediction on samples from the simulation and real-world training splits of RoboDojo [72]. We also include real-world data: self-collected observations and samples from the dataset released with RDT-1B [82]. Given the current observation and task instruction, the understanding expert identifies a task-relevant object or region and predicts its bounding box. Fig. 10 shows the predicted target names and bounding boxes in this setting. These examples include objects such as a charger, fruits, a pen, and a straw, as well as regions inside a bowl and on a table. These qualitative results suggest that ME-U0 can identify and localize task-relevant objects and regions in both simulated and real-world scenes.

7.4.3 Subtask

We evaluate subtask prediction on the same samples used for affordance prediction. Given the overall task instruction and current observation, the understanding expert describes what the robot should do at the current stage of the task. Fig. 10 also demonstrates the predicted subtasks and their corresponding affordance targets. Each subtask specifies the action to perform at the current stage, while the corresponding affordance target identifies the task-relevant object or region.

8 Conclusion

We presented **MachEmbodied-U0 (ME-U0)**, a unified embodied foundation model that connects task-grounded understanding, geometry- and motion-aware visual dynamics, and continuous action generation. Through a Mixture-of-Transformers architecture, subtask prediction and affordance grounding provide the generation process with semantic and spatial context, while future RGB, depth, surface normals, optical flow, and continuous actions are jointly modeled through flow matching. Pretraining on approximately 4,200 hours of demonstrations across six robot embodiments enables ME-U0 to achieve competitive performance on RoboDojo, LIBERO, and LIBERO-Plus, with additional validation on real-world manipulation tasks. The model also retains zero-shot subtask prediction, affordance grounding, and visual-dynamics generation without corresponding downstream supervision.

Future work will extend ME-U0 to broader distributions of tasks, environments, and embodiments, while strengthening long-term memory and in-context adaptation for more general embodied interaction.

9 Contributions and Acknowledgments

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